

MSML640 Final Project Proposal

Traffic Sign Classification with Transfer Learning: Studying the Effect of Augmentation and Synthetic Data

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1. Project Idea

For the final project, we plan to build an image classification system for recognizing traffic signs using transfer learning. The main goal is to study how a pretrained deep learning model can be adapted to a new classification task and how different data preparation strategies affect performance. In particular, we will compare a baseline model trained on the original dataset only against versions trained with augmentation, synthesized data, and a combination of both. The project will focus not only on classification accuracy, but also on robustness under realistic distribution shifts such as blur, noise, partial occlusion, and lighting changes. Traffic signs are a useful benchmark for this problem because even small visual distortions can significantly affect recognition quality.

2. Dataset

We will use the **German Traffic Sign Recognition Benchmark (GTSRB)**, a real-world public dataset for traffic sign classification. The full dataset contains 43 classes, but for this project we will use a smaller subset of 6 visually distinct classes so that the experiments remain manageable and the analysis is easier to interpret.

The selected classes are:

- 00012 — Priority road
- 00013 — Yield
- 00014 — Stop
- 00017 — No entry
- 00025 — Road work
- 00038 — Keep right

These classes were chosen because they are visually diverse in terms of shape, color, and symbol, which makes them suitable for a clean classification study. The dataset includes separate training and test images, and we will further split the training portion into training and validation sets with no overlap between splits.

3. Model / Backbone Choice

We will use **ResNet50 pretrained on ImageNet** as the backbone model.

This choice is appropriate because:

- **ResNet50** is a standard and well-established transfer learning backbone

- it provides strong performance on image classification tasks
- it is expressive enough for this dataset while still being practical to fine-tune
- it is easy to justify and explain in the final report and presentation

The final classification layer will be replaced so that the network predicts the 6 selected traffic sign classes.

4. Planned Training Configurations

We will fine-tune the model under the four required configurations:

1. **Baseline**
Original data only, with no augmentation or synthesis.
2. **Original data + augmentation**
Standard augmentations such as mild rotation, crop, translation, brightness adjustment, and color jitter applied to the training set.
3. **Original data + synthesized data**
Additional synthetic training samples will be created from the original traffic sign images. These will simulate realistic variations such as partial occlusion, altered backgrounds, brightness changes, and other controlled perturbations.
4. **Original data + synthesized data + augmentation**
A combination of both synthesized training samples and standard training augmentation.

5. Evaluation Plan

For all four configurations, we will report:

- training and validation loss curves
- training and validation accuracy per epoch
- confusion matrices on the test set

In addition, we will provide a focused comparison between the baseline model and the best-performing model using:

- side-by-side confusion matrices
- error analysis of misclassified examples
- discussion of which classes are confused and why
- robustness testing under noise, blur, occlusion, and lighting changes

6. Expected Application

The final application will be a traffic sign classifier that takes an input image and predicts the traffic sign category among the selected classes. Beyond simply making predictions, the project will demonstrate how different data strategies influence classification quality and robustness. This makes the project meaningful because it connects transfer learning with practical issues in real-world recognition systems, where image quality and viewing conditions are often imperfect. The originality of the project will come from the focused comparison of augmentation and synthesized data, along with robustness testing under multiple controlled perturbations.